Case name: HORUS

Description: 5kn liquid rocket engine

Tue Sep 22 21:04:47 2026

Table 1. Propellant Specification

Component	Temperature, K	Pressure, MPa	Enthalpy, kJ/mol	Enthalpy, kJ/kg	Mass fraction
C4H10O(L)	298.1	0.1013	-328.0000	-4425.161	0.2500000
N2O(L),298.15K	298.1	0.1013	65.1000	1479.115	0.7500000
Total			0.1492	3.046	

Propellant exploded formula:

$C_{0.660908}\ H_{1.652271}\ O_{1.000000}\ N_{1.669546}$ (based on 1 mole)

$C_{13.263995}\ H_{33.159987}\ O_{20.069338}\ N_{33.506679}$ (by mass %)

α: 0.4210197 (oxidizer excess coefficient)

O/F: 3.0000000

O/F_0: 7.1255569 (stoichiometric)

rho: kg/m^3

Table 2. Combustion Properties

Parameter	Injector	Nozzle inlet	Nozzle throat	A/At=1.500	A/At=2.000	A/At=2.500	A/At=3.000	A/At=3.500	A/At=4.000	A/At=4.500	Nozzle exit	Unit
Pressure	3.5000	3.4389	1.9098	0.6230	0.3781	0.2639	0.1989	0.1576	0.1293	0.1090	0.0977	MPa
Temperature	2240.4782	2236.2996	1973.8701	1546.7921	1385.4628	1279.6890	1202.5407	1143.0416	1095.6123	1057.1778	1034.1875	K
Enthalpy	64.0733	-101.0788	-10385.7924	-26710.4229	-32793.3049	-36777.6808	-39697.0167	-41973.5823	-43827.3796	-45385.8477	-46362.8659	J/mol
	3.0376	-4.7920	-492.2639	-1265.9210	-1554.1864	-1742.9488	-1881.1321	-1988.6512	-2075.7805	-2148.3687	-2193.3348	kJ/kg
Entropy	228.3459	228.4195	228.4696	228.4867	228.4909	228.5002	228.5206	228.5621	228.6393	228.7697	228.9034	J/(mol·K)
	10.8255	10.8290	10.8290	10.8290	10.8290	10.8290	10.8290	10.8290	10.8290	10.8290	10.8290	kJ/(kg·K)
Internal energy	-18564.3203	-18694.7296	-26797.4802	-39571.1826	-44312.6966	-47417.6191	-49695.5080	-51477.3700	-52936.8176	-54175.7233	-54961.5887	J/mol
	-880.1049	-886.2839	-1270.1421	-1875.4474	-2100.1297	-2247.1913	-2354.9330	-2438.9277	-2507.2275	-2564.4432	-2600.1232	kJ/kg
Specific heat (p=const)	1.8891	1.8883	1.8390	1.7917	1.7851	1.7894	1.8044	1.8352	1.8924	1.9898	2.0908	kJ/(kg·K)
Specific heat (V=const)	1.4918	1.4910	1.4440	1.3974	1.3906	1.3942	1.4075	1.4348	1.4850	1.5703	1.6586	kJ/(kg·K)
Gamma	1.2663	1.2664	1.2736	1.2822	1.2837	1.2834	1.2819	1.2791	1.2743	1.2671	1.2606	
Isentropic exponent	1.2661	1.2662	1.2735	1.2821	1.2836	1.2833	1.2815	1.2782	1.2726	1.2640	1.2561	
Gas constant	0.3942	0.3942	0.3941	0.3941	0.3941	0.3940	0.3940	0.3939	0.3938	0.3936	0.3933	kJ/(kg·K)
Molecular	21.0933	21.0934	21.0980	21.0996	21.1000	21.1008	21.1027	21.1066	21.1137	21.1257	21.1381	

Parameter	Injector	Nozzle inlet	Nozzle throat	A/At=1.500	A/At=2.000	A/At=2.500	A/At=3.000	A/At=3.500	A/At=4.000	A/At=4.500	Nozzle exit	Unit
weight												
Density	3.9631	3.9012	2.4551	1.0221	0.6925	0.5233	0.4198	0.3500	0.2997	0.2619	0.2401	kg/m^3
Sonic velocity	1057.4062	1056.4635	995.2901	884.0083	837.1232	804.4069	779.2211	758.6405	740.9779	725.2104	714.8270	m/s
Mach number	0.0000	0.1184	1.0000	1.8021	2.1081	2.3231	2.4912	2.6308	2.7518	2.8603	2.9320	
Area ratio	5.0000	5.0000	1.0000	1.5000	2.0000	2.5000	3.0000	3.5000	4.0000	4.5000	4.8572	A/At
Mass flux	488.1841	488.1841	2443.5172	1628.2282	1222.0996	977.8104	814.8470	698.4813	611.1599	543.1975	503.2492	$kg/(m^2 \cdot s)$
Viscosity	0.7247	0.7238	0.6643	0.5631	0.5230	0.4959	0.4756	0.4596	0.4466	0.4359	0.4294	$x10^{-4}$ kg/(m·s)
Conductivity, frozen	0.2293	0.2289	0.2057	0.1678	0.1537	0.1446	0.1380	0.1330	0.1291	0.1259	0.1240	W/(m·K)
Specific heat (p=const), frozen	1.8374	1.8370	1.8064	1.7415	1.7110	1.6893	1.6726	1.6593	1.6485	1.6397	1.6344	kJ/(kg·K)
Prandtl number, frozen	0.5807	0.5808	0.5833	0.5843	0.5822	0.5794	0.5763	0.5732	0.5703	0.5676	0.5659	
Conductivity, effective	0.2473	0.2467	0.2124	0.1682	0.1537	0.1446	0.1381	0.1331	0.1291	0.1259	0.1240	W/(m·K)
Specific heat (p=const), effective	1.8891	1.8883	1.8389	1.7432	1.7112	1.6896	1.6730	1.6599	1.6495	1.6397	nan	kJ/(kg·K)
Prandtl number, effective	0.5536	0.5540	0.5750	0.5837	0.5821	0.5794	0.5763	0.5733	0.5705	0.5676	nan	

Table 3. Combustion Products

Product	Injector mass fraction	Injector mole fraction	Nozzle inlet mass fraction	Nozzle inlet mole fraction	Nozzle throat mass fraction	Nozzle throat mole fraction	A/At=1.500 mass fraction	A/At=1.500 mole fraction	A/At=2.000 mass fraction	A/At=2.000 mole fraction	A/At=2.500 mass fraction	A/At=2.500 mole fraction	A/At=3.000 mass fraction	A/At=3.000 mole fraction	A/At=3.500 mass fraction	A/At=3.500 mole fraction	A/At=4.000 mass fraction	A/At=4.000 mole fraction	A/At=4.500 mass fraction	A/At=4.500 mole fraction	Nozzle exit mass fraction	Nozzle exit mole fraction
CH4	0.00000003	0.00000004	0.00000003	0.00000004	0.00000005	0.00000006	0.00000027	0.00000035	0.00000085	0.00000112	0.00000232	0.00000304	0.00000563	0.00000740	0.00001243	0.00001636	0.00002517	0.00003312	0.00004672	0.00006153	0.00006884	0.00009070
CO	0.34490	0.25973	0.34484	0.25968	0.34032	0.25634	0.32692	0.24626	0.31811	0.23963	0.31038	0.23382	0.30340	0.22858	0.29699	0.22379	0.29103	0.21937	0.28541	0.21526	0.28158	0.21249
CO2	0.05180	0.02482	0.05189	0.02487	0.05900	0.02828	0.08007	0.03839	0.09389	0.04501	0.10599	0.05082	0.11687	0.05604	0.12676	0.06079	0.13578	0.06514	0.14402	0.06913	0.14943	0.07177

Product	Injector mass fraction	Injector mole fraction	Nozzle inlet mass fraction	Nozzle inlet mole fraction	Nozzle throat mass fraction	Nozzle throat mole fraction	A/At=1.500 mass fraction	A/At=1.500 mole fraction	A/At=2.000 mass fraction	A/At=2.000 mole fraction	A/At=2.500 mass fraction	A/At=2.500 mole fraction	A/At=3.000 mass fraction	A/At=3.000 mole fraction	A/At=3.500 mass fraction	A/At=3.500 mole fraction	A/At=4.000 mass fraction	A/At=4.000 mole fraction	A/At=4.500 mass fraction	A/At=4.500 mole fraction	Nozzle exit mass fraction	Nozzle exit mole fraction
	21	81	43	24	83	82	66	11	78	83	68	11	92	38	49	50	84	45	20	40	63	49
CO	0.00000002	0.00000001	0.00000002	0.00000001																		
OH	0.00000276	0.00005779	0.00000272	0.00005699	0.00000073	0.00001518	0.00000003	0.00000061														
H2	0.02238420	0.23491834	0.02238080	0.23425885	0.02273232	0.23792020	0.02370250	0.24811414	0.02433460	0.25472012	0.02488045	0.26025537	0.02557298	0.26579070	0.02616804	0.27009068	0.02645019	0.27401004	0.02675837	0.27726075	0.02662019	0.27914397
H2O	0.10346081	0.12113713	0.10342391	0.12109434	0.10054848	0.11077492	0.09192705	0.10766491	0.08627534	0.10104758	0.08133886	0.09526951	0.07692113	0.09010332	0.07295078	0.08546809	0.06939985	0.08133541	0.06627144	0.07771324	0.06430345	0.07544954
HCHO,formaldehy	0.00000012	0.00000009	0.00000012	0.00000009	0.00000007	0.00000005	0.00000002	0.00000002	0.00000002	0.00000001	0.00000001	0.00000001										
HCN	0.00001025	0.00000798	0.00001023	0.00000796	0.00000072	0.00000057	0.00000028	0.00000022	0.00000019	0.00000015	0.00000015	0.00000012	0.00000012	0.00000010	0.00000011	0.00000008	0.00000009	0.00000007	0.00000008	0.00000007	0.00000008	0.00000006
HCO	0.00000016	0.00000011	0.00000015	0.00000011	0.00000004	0.00000003																
HCOOH	0.00000009	0.00000004	0.00000009	0.00000004	0.00000005	0.00000002	0.00000002	0.00000001	0.00000001	0.00000001												
HNCO	0.00000009	0.00000007	0.00000009	0.00000007	0.00000003	0.00000003																
HNCO	0.00000029	0.00000014	0.00000029	0.00000014	0.00000016	0.00000008	0.00000006	0.00000003	0.00000004	0.00000002	0.00000003	0.00000001	0.00000002	0.00000001	0.00000002	0.00000001	0.00000001	0.00000001	0.00000001	0.00000001	0.00000001	0.00000001
N2	0.47733000	0.35941430	0.47733040	0.35941600	0.47733890	0.35950130	0.47734390	0.35953200	0.47734360	0.35953830	0.47734250	0.35955210	0.47734100	0.35958310	0.47733930	0.35964710	0.47733760	0.35976730	0.47733600	0.35997120	0.47733490	0.36018080
NH2	0.00000002	0.00000003	0.00000002	0.00000003																		
NH3	0.00000272	0.00000337	0.00000269	0.00000333	0.00000227	0.00000281	0.00000202	0.00000251	0.00000213	0.00000263	0.00000230	0.00000285	0.00000250	0.00000310	0.00000271	0.00000336	0.00000293	0.00000364	0.00000314	0.00000389	0.00000327	0.00000406
NO	0.00000037	0.00000026	0.00000036	0.00000025	0.00000004	0.00000003																
OH	0.00000398	0.00000493	0.00000390	0.00000483	0.00000065	0.00000081																

Product	Injector mass fraction	Injector mole fraction	Nozzle inlet mass fraction	Nozzle inlet mole fraction	Nozzle throat mass fraction	Nozzle throat mole fraction	A/At= 1.500 mass fraction	A/At= 1.500 mole fraction	A/At= 2.000 mass fraction	A/At= 2.000 mole fraction	A/At= 2.500 mass fraction	A/At= 2.500 mole fraction	A/At= 3.000 mass fraction	A/At= 3.000 mole fraction	A/At= 3.500 mass fraction	A/At= 3.500 mole fraction	A/At= 4.000 mass fraction	A/At= 4.000 mole fraction	A/At= 4.500 mass fraction	A/At= 4.500 mole fraction	Nozzle exit mass fraction	Nozzle exit mole fraction
Gaseous fraction:	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000	1.0000000
Condensed fraction:	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000	0.0000000

Table 4. Ideal Performance

Parameter	Sea level	Optimum expansion	Vacuum	Units
Characteristic velocity		1419.9024		m/s
Effective exhaust velocity	2088.6358	2095.8876	2289.9774	m/s
Specific impulse (by mass)	2088.6358	2095.8876	2289.9774	N·s/kg
Specific impulse (by weight)	212.9816	213.7211	233.5127	s
Thrust coefficient	1.4710	1.4761	1.6128	
Thrust	4.1781	4.1926	4.5809	kN
Altitude	0.0000	0.3080		km
Ambient pressure	1.0000	0.9640	0.0000	atm

Table 5. Estimated Delivered Performance

Parameter	Sea level	Optimum expansion	Vacuum	Units
Characteristic velocity		1381.3330		m/s
Effective exhaust velocity	1999.5952	2006.6500	2195.4677	m/s
Specific impulse (by mass)	1999.5952	2006.6500	2195.4677	N·s/kg
Specific impulse (by weight)	203.9020	204.6214	223.8754	s
Thrust coefficient	1.4476	1.4527	1.5894	
Thrust	4.0000	4.0141	4.3918	kN
Altitude	0.0000	0.3080		km
Ambient pressure	1.0000	0.9640	0.0000	atm

Table 6. Altitude Performance

Altitude, km	Pressure, atm	Specific impulse, N·s/kg	Specific impulse, s	Thrust coefficient	Thrust, kN
0.0000	1.0000	1999.5952	203.9020	1.4476	4.0000

Table 7. Throttled Performance

Table 8. Chamber Size

Combustion chamber size			Nozzle size		
Dc	71.21	mm	Type	Parabolic nozzle	
Dt	31.84	mm	Rn	6.08	mm
Lcyl	217.99	mm	Tn	21.36	deg
Lc	286.13	mm	Te	14.07	deg
L*	1306.40	mm	De	70.18	mm
R1	23.88	mm	Le	57.87	mm
R2	93.09	mm	Le/Dt	1.82	
b	25.00	deg	Le/Lc15	80.00	%
Ac/At	5.00		Ae/At	4.86	

Parameter		**Engine**	**Chamber**	
Thrust	sea level	4.0000	4.0000	kN
	opt exp	4.0141	4.0141	kN
	vacuum	4.3918	4.3918	kN
Specific Impulse	sea level	1999.5952	1999.5952	N·s/kg
	opt exp	2006.6500	2006.6500	N·s/kg
	vacuun	2195.4677	2195.4677	N·s/kg
Mass flow rate	total	2.0004	2.0004	kg/s
	oxidizer	1.5003	1.5003	kg/s
	fuel	0.5001	0.5001	kg/s
	film coolant	n.a.	n.a.	kg/s
Number of chambers		1		

Table 10. Thermal Analysis